

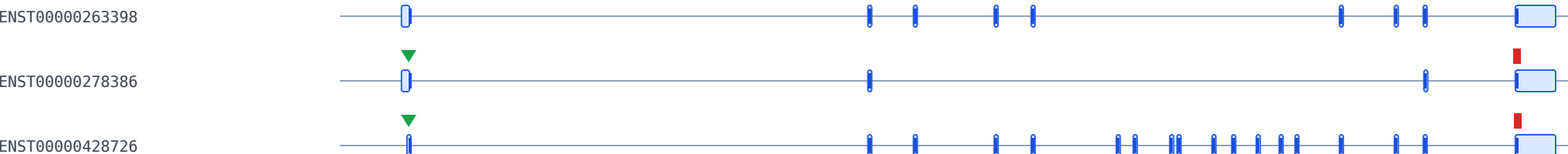
CD44 promoter-reporter architectures

5' -> 3' display, genomic 35133721 -> 35233402 | strand + | assembly GRCh38 | annotation Ensembl live GRCh38; selected reporter transcripts | upstream 5000 bp | downstream 1000 bp

Canonical reporter comparison: 2 TSS class(es), 3 transcript(s), 9 explicit architecture row(s); normalized evidence remains observational/predictive context.



Transcript models and annotation-derived metrics

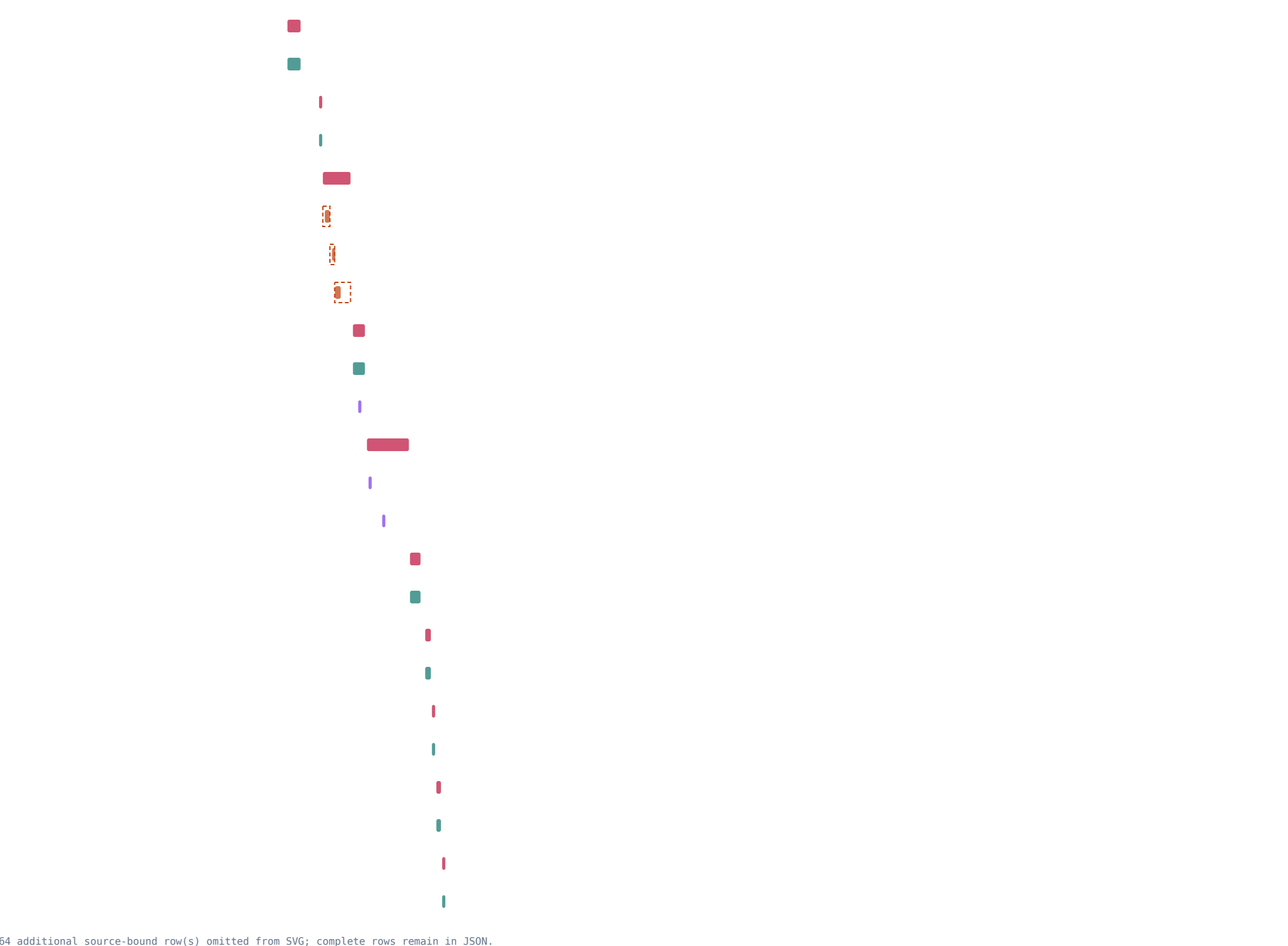


Proposed reporter architectures



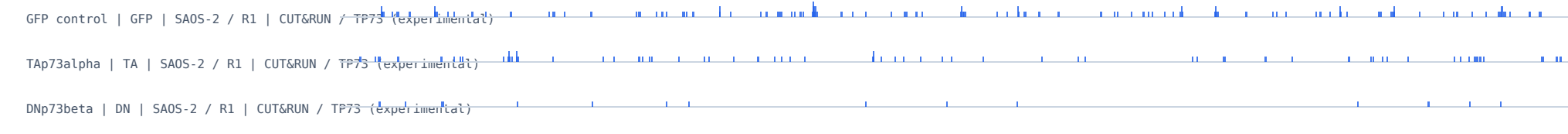
Ensembl-annotated regulatory regions

Ensembl Regulation | release 2026-08 | pipeline 2.1 | GRCh38

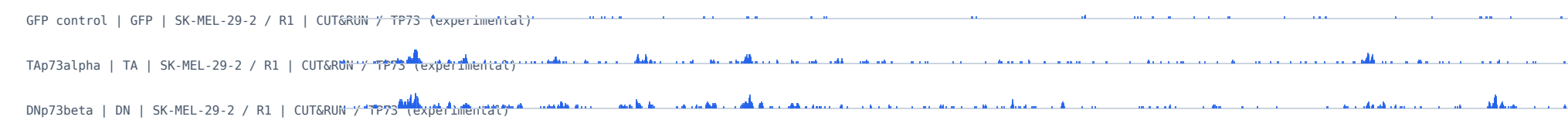


Projected occupancy grouped by declared biological context

TP73 occupancy — SAOS-2 | shared within group | observed group max 3.000



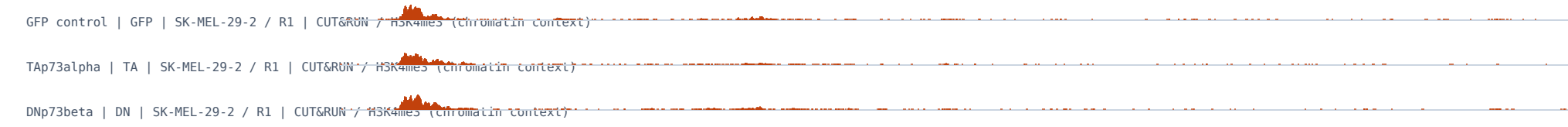
TP73 occupancy — SK-MEL-29-2 | shared within group | observed group max 9.000



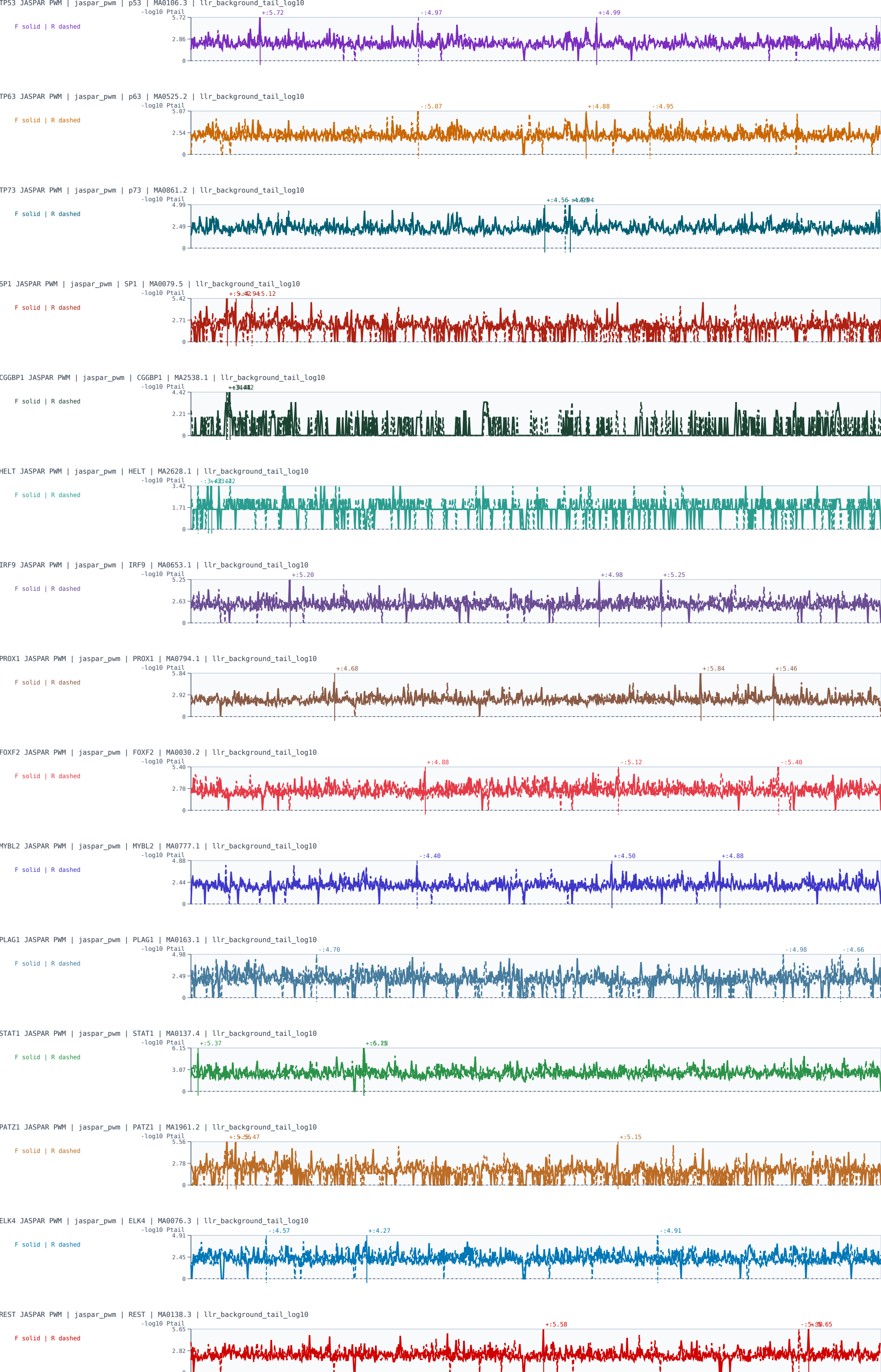
H3K4me3 chromatin context — SAOS-2 | shared within group | observed group max 6.000



H3K4me3 chromatin context — SK-MEL-29-2 | shared within group | observed group max 44.000



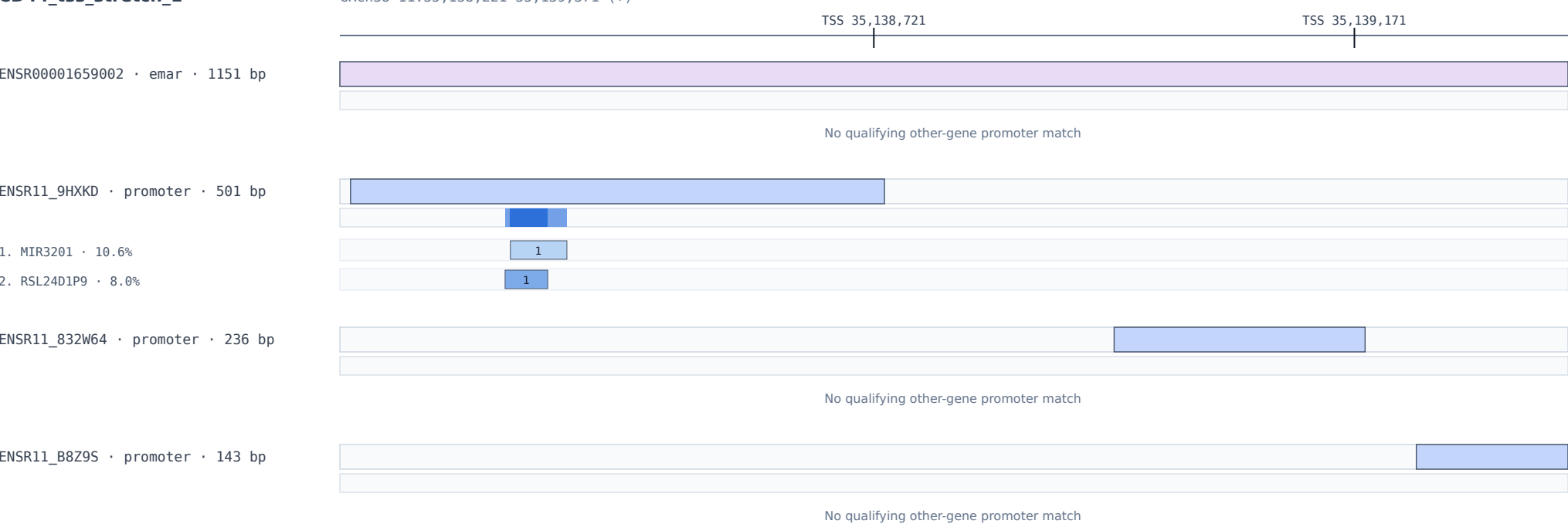
Predicted TF binding scores



TSS-local Ensembl-feature promoterome similarity

Transcript-oriented -500/+200 bp; features clipped to the displayed stretch; BLASTN ≥40 bp, ≥80% identity, E=1e-05; self-locus and same-gene targets excluded. Blue frequency strips count distinct other genes by query position. Numbered blocks give target-priority order; red outlines mark order/orientation breaks.

CD44_tss_stretch_1



Reporter interpretation boundaries

Primary interpretation boundaries: Ensembl-annotated TSS and CUT&RUN and ChIP-seq evidence. CUT&RUN overlap is occupancy/enrichment evidence, not proof of TSS usage, promoter activity, direct regulation, biochemical affinity, or luciferase output. Reporter geometry does not verify the identity of a physical laboratory construct.

Warnings / interpretation checks

Evidence provenance

ensembl_regulation_annotation_index: ensembl_regulation_2026_08_grch38
ensembl_regulation_annotation_intervals: ensembl_regulation_2026_08_grch38
isoform_panel: CD44_panel
occupancy_layout: tp73_saos2,tp73_skmel29_2,h3k4me3_saos2,h3k4me3_skmel29_2
predicted_tf_score_jaspar_pwm: local_jaspar_registry
projected_occupancy_track: E-MTAB-15789:h3k4me3:saos2:DN:R1
projected_occupancy_track: E-MTAB-15789:h3k4me3:saos2:GFP:R1
projected_occupancy_track: E-MTAB-15789:h3k4me3:saos2:TA:R1
projected_occupancy_track: E-MTAB-15789:h3k4me3:skmel29-2:DN:R1
projected_occupancy_track: E-MTAB-15789:h3k4me3:skmel29-2:GFP:R1